Definition

- The function f is called **invertible/ bijective** (or a **one-to-one correspondence**, or a **bijection**), if it is both injective and surjective.
- Let $f : A \to B$ be a bijection. The **inverse function** of f is the function that assigns to every $b \in B$ the unique element $a \in A$ such that $f(a) = b$.
- The inverse function of f is denoted by f^{-1}.

bijective: injective and surjective
not bijective: not injective or not surjective.

Example. Determine whether each of the following functions is bijective or not. If yes, find its inverse.

1. $f : \mathbb{N} \to \mathbb{N}$, $f(n) = n + 1$

Check f is injective: if f(a) = f(b), then a + 1 = b + 1, then —1 both sides, we get a = b.

Check f is not surjective: because f(n) = n+1 = 1, then n = 0 which is not in the domain N, so 1 is not in the image.

So f is not bijective.

Example. Determine whether each of the following functions is bijective or not. If yes, find its inverse.

1. $f : \mathbb{N} \to \mathbb{N}$, $f(n) = n + 1$

2. $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x - 1$

Check f is injective: f(x) = f(y), then 2x —1 = 2y —1, then 2x = 2y, then x = y.

Check f is not surjective: because we can find even numbers in the codomain Z, these are not coming from the domain Z via the function f(x) = 2x — 1 (always odd).

f is not bijective.

Example. Determine whether each of the following functions is bijective or not. If yes, find its inverse.

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Check f is not surjective: Given b in codomain Z, We solve for a in the domain Z such that f(a) = 2a — 1 = b, ……, a = (b+1) / 2. This a is not in the domain Z if b is en even number,

3. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x - 1$

Check f is injective: (the same as the last question)

Check f is surjective: Given b in codomain R, We solve for a in the domain R such that f(a) = 2a — 1 = b, ……, a = (b+1) / 2. This a is a real number, so is in the domain R.

So f is bijective, the inverse: f^{-1} : R —> R, f^{-1}(b) = (b+1) / 2.

Example. Determine whether each of the following functions is bijective or not. If yes, find its inverse.

1. $f : \mathbb{N} \to \mathbb{N}$, $f(n) = n + 1$

2. $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = 2x - 1$

3. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 2x - 1$

Remember: cubic-root $x \to x^{1/3}$ can be taken on the whole R.

4. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^3 + 1$.

Check f is injective: f(x) = f(y), then x^3 + 1 = y^3 + 1, then x^3 = y^3, then x = y.

Check f is surjective: given b in R, we solve for a in the equation: a^3 + 1 = b, …

$a = (b-1)^{1/3}$ ← $f^{-1} : R \to R$, $f^{-1}(b) = (b-1)^{1/3}$

Example. Determine whether each of the following functions is bijective or not. If yes, find its inverse.

1. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

f is not injective: because f(—2) = f(2) but —2 is not = 2.

f is not surjective: because —4 is not an image of f.

f is not bijective (either because it is not injective or not surjective)

2. $f : [0, \infty) \to [0, \infty)$, $f(x) = x^2$

f is surjective: for each y in [0,∞), there is one x in [0,∞), such that x^2 = y.
f is injective: the x we found above is unique in [0,∞).

f is bijective: f^{-1} : [0,∞) →[0,∞), $f^{-1}(y) = y^{1/2}$ square-root can be taken on non-negative numbers

3. $f : (-\infty, 0] \to [0, \infty)$, $f(x) = x^2$

The work is almost the same as above.

f is surjective: for each y in [0,∞), there is one x in (–∞,0] , such that x^2 = y.
f is injective: the x we found above is unique in (–∞,0] .

f is bijective: f^{-1} : [0,∞) —> (–∞,0] , $f^{-1}(y) = -\, y^{1/2}$ By default, a square-root is non-negative, so we put a — sign to get a non-positive output.

Beware of notations

- f^{-1} can be defined only when f is invertible.
  That means: if f is not invertible, we cannot write f^{-1}.
- However, if T is a subset of B, we can always write the preimage
  $$f^{-1}(T) = \{a \in A | f(a) \in T\},$$
  even though f is not invertible.
- The two notations are consistent: when f is invertible, let $g = f^{-1}$, then the image $g(T)$, which is
  $$g(T) = \{g(b) | b \in T\},$$
  is equal to the preimage $f^{-1}(T)$.

2.5 Cardinality of Sets

Definition

- Two sets A and B have the same **cardinality** if and only if there exists a bijection from A to B.
- In this case, we write $|A| = |B|$.

Remark. A and B can be infinite sets!

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Remark. A and B can be infinite sets!

Example. Let $\mathbb{O}$ be the set of odd positive integers.
Let $\mathbb{N}$ be the set of positive integers.

We see that O is a <u>proper</u> subset of N. However, we can establish a bijection

f: N —> O, f(n) = 2n—1. This function is injective (checked before)
f is also surjective: for every b in O, so b is an odd positive integer, so
b must be in the form 2a —1 for some positive integer a, and so f(a) = b.

f is a bijection from N to O, so we have INI = IOI.

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Remark. A and B can be infinite sets!

Example. Let $\mathbb{O}$ be the set of odd positive integers.
Let $\mathbb{N}$ be the set of positive integers.
The map

$$f : \mathbb{N} \to \mathbb{O}, \quad n \mapsto 2n - 1$$

is a bijection, so $|\mathbb{O}| = |\mathbb{N}|$.

Example. $\mathbb{N} \setminus \{1\} = \{2, 3, 4, 5, \cdots\} \subset \mathbb{N}$, but $|\mathbb{N} \setminus \{1\}| = |\mathbb{N}|$.

The function f: N \{1} —> N, f(n) = n—1, is a bijection.
(The checking of injective and surjective are easy.)

Example

- $2\mathbb{Z} \subsetneq \mathbb{Z}$ but $|2\mathbb{Z}| = |\mathbb{Z}|$

  f: Z —> 2Z, f(a) = 2a is a bijection.

  can also use g: 2Z —> Z, g(a) = a/2, also a bijection.

Example

- $2\mathbb{Z} \subsetneq \mathbb{Z}$ but $|2\mathbb{Z}| = |\mathbb{Z}|$

- $(0, \infty) \subsetneq \mathbb{R}$ but $|(0, \infty)| = |\mathbb{R}|$

  f: (0,∞) —> R, f(x) = log(x), refer to the graph on the board.

Example

- $2\mathbb{Z} \subsetneq \mathbb{Z}$ but $|2\mathbb{Z}| = |\mathbb{Z}|$

- $(0, \infty) \subsetneq \mathbb{R}$ but $|(0, \infty)| = |\mathbb{R}|$

- $(-\pi/2, \pi/2) \subsetneq \mathbb{R}$ but $|(\pi/2, \pi/2)| = |\mathbb{R}|$

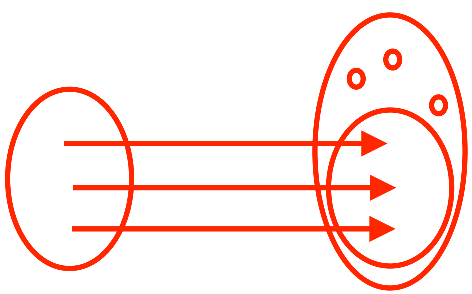
an interval of finite length

f: (–π / 2, π /2) —> R, f(x) = tan(x).

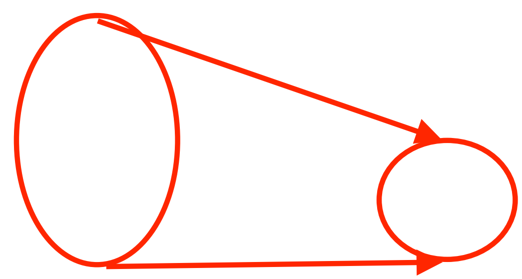
Definition

- If there is only an injective function from A to B, the cardinality of A is less than the cardinality of B and we write $|A| \leq |B|$ or $|B| \geq |A|$. or equal to
- If there is only a surjective function from A to B, the cardinality of A is more than the cardinality of B and we write $|A| \geq |B|$ or $|B| \leq |A|$. or equal to

injection: smaller set —> larger set



surjection: larger set —> smaller set



Definition: Countability

- A set that is either finite or has the same cardinality as the set of positive integers $\mathbb{N}$ is called **countable**.

counting set

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Definition: Countability

- A set that is either finite or has the same cardinality as the set of positive integers $\mathbb{N}$ is called **countable**.
- Usually we emphasize that $\mathbb{N}$ is **infinitely countable**.
- A set that is not countable is called **uncountable**.

Example. The set $\mathbb{O} = 2\mathbb{N} - 1$ of odd positive integers is infinitely countable. f: N —> O, f(n) = 2n—1 is a bijection.

O = { 1, 3, 5, 7, 9, …}

… 1-1 correspondence/bijection

N = { 1, 2, 3, 4, 5, … }

Theorem

The set $\mathbb{Z}$ of all integers is infinitely countable.

Jump counting:

Z = {…, —3 , —2, —1, 0, 1, 2, 3, …}

bijective

N = { 1, 2, 3, 4, 5, 6, … }

f: N —> Z, f(n) = n/2 when n is even, f(n) = — (n—1)/2 when n is odd.

Theorem

The set $\mathbb{Z}$ of all integers is infinitely countable.

Example. $S = \{2^k | k \in \mathbb{Z}\}$

$$= \{\dots, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8, 16, 32, 64, \dots\}.$$

We already showed that Z is infinitely countable from above.

g: Z —> S, g(k) = 2^k is a bijection between S and Z, so S is infinitely countable.

A bijection between S and N is the composition g o f: $N \xrightarrow{f} Z \xrightarrow{g} S$

(g o f)(n)

Assume all sets are infinite (if the sets are finite, the proofs are easy).

Some simple properties.

- If $|A| \leq |B|$, and B is countable, then A is countable.

|A|≤ |B| means: there is an injection f: A —> B,
and B countable means: there is a bijection g: B —> N (set of natural numbers)
then the composition g o f: A —> N is also an injection, then |A| < |N| or |A| = |N|.
The first case means A is finite, the second case means A is infinitely countable.
In both cases A is countable.

Some simple properties.

- If $|A| \leq |B|$, and B is countable, then A is countable.

- If $|A| \leq |B|$, and A is uncountable, then B is uncountable.

Use contradiction (assume the conclusion is wrong, and derive the assumption is also wrong)
“Assume B is countable”, given |A| ≤ |B|, the first statement above implies A is countable.
This is a contradiction to the assumption that A is uncountable.

So “Assume B is countable” is wrong, that means B is uncountable.

Some simple properties.

- If $|A| \leq |B|$, and B is countable, then A is countable.

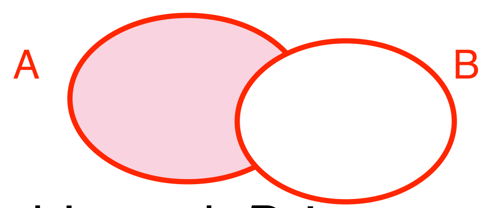
- If $|A| \leq |B|$, and A is uncountable, then B is uncountable.

- If A and B are countable sets, then $A \cup B$ and $A \cap B$ are also countable.

To show A ∪ B is counable, use jump counting method from last page.
To show A ∩B is countable, just observe we have an injection A ∩B —> A (or B) mapping x in A ∩B to itself, which is in A because A ∩B is a subset of A.
I A ∩B I ≤ IAI, then apply the first statement above.

Some simple properties.

- If $|A| \leq |B|$, and B is countable, then A is countable.
- If $|A| \leq |B|$, and A is uncountable, then B is uncountable.
- If A and B are countable sets, then $A \cup B$ and $A \cap B$ are also countable.



- If A is uncountable and B is countable, then $A \setminus B$ is uncountable.

Use contradiction: if A \ B is countable, then A is a subset of (A\B) ∪B = A ∪B.
Given both A \ B and B countable, we have A ∪B also countable, and so A is also countable.
But A is given to be uncountable, this is a contradiction.
So “A \ B countable” is wrong, so A \ B is uncountable.

Some simple properties. Let $f : A \to B$ be a function.

- If A is countable, then the image $f(A)$ is also countable.

If A is (infinitely) countable, then there is a bijection g: N —> A

If we view f : A —> f(A) (we view the image as the new codomain), then f is surjective.

then we have a surjection f o g: N —> A —> f(A), then |N| ≥ |f(A)|, so f(A) is also countable.

Some simple properties. Let $f : A \to B$ be a function.

- If A is countable, then the image $f(A)$ is also countable.

- In particular, if f is <u>surjective</u> and A is countable, then B is countable.

  f(A) = B

Theorem

If A and B are countable sets, then

$$A \times B \text{ is also countable.}$$

Cartesian product A ×B = { (a,b) l a in A and b in B }

Theorem

If A and B are countable sets, then

$$A \times B \text{ is also countable.}$$

Proof. Diagonal counting:

$$\begin{array}{ccccc}
(a_1, b_1) & (a_1, b_2) & (a_1, b_3) & (a_1, b_4) & \dots \\
(a_2, b_1) & (a_2, b_2) & (a_2, b_3) & (a_2, b_4) & \dots \\
(a_3, b_1) & (a_3, b_2) & (a_3, b_3) & (a_3, b_4) & \dots \\
(a_4, b_1) & (a_4, b_2) & (a_4, b_3) & (a_4, b_4) & \dots \\
\vdots & \vdots & \vdots & \vdots & \ddots
\end{array}$$

We can cover the whole A x B using these diagonals.

This give a bijection from N —> A x B (although the exact formula is difficult to write down).

N (positive integers) ⊂ Z (all integers) ⊂ Q (rational numbers) ⊂ R (real numbers)

(infinitely) countable   countable

Theorem. The set $\mathbb{Q}$ of rational numbers is countable.

Theorem. The set $\mathbb{Q}$ of rational numbers is countable.

Proof. There is a surjective map $\mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$.

f: Z ×N →Q, f(a,b) = a/b. It is surjective because any fraction is given by a/b.

Z and N are countable, so the Cartesian product Z ×N is also countable
(by the theorem on last page).

f is surjection, Z ×N is also countable, then the image (which is Q) is also countable
(by the second fact on p.35).

Theorem. The set $\mathbb{R}$ of real numbers is uncountable.

Proof by contradiction: Suppose that $\mathbb{R}$ is countable,

$$\mathbb{R} = \{r_1, r_2, r_3, \ldots\}, \quad \textit{where}$$

$$r_1 = n_1.d_{11}d_{12}d_{13}\ldots$$

(decimal expressions)

pi = 3. 1 4 1 6

$$r_2 = n_2.d_{21}d_{22}d_{23}\ldots$$

sqrt(2) = 1. 4 1 4 …

$$r_3 = n_3.d_{31}d_{32}d_{33}\ldots$$

Here each n_i is the integer part of r_i, and $d_{ij} \in \{0, 1, \ldots, 9\}$ are the digits of r_i.

i means the i-th real number
j means the j-th decimal place of ri.

Construct a new number:

r = 0. e1 e2 e3 …, where ei is not equal to d_ii

The i-th decimal digit of r is different from d_ii, the i-th decimal digit of r_i.
That means: r is different from all r_i .

This r is a real number, but r is not equal to any number in the list {r1, r2, r3, …}

This is a contradiction to: "R is countable", so R must be uncountable.